Solution for 2023 Fall Midterm Exam (30 points)

1. (4 points) Newtonian mechanics with Galilean relativity is good enough for describing all phenomena with $v << c$. What was the original reason of developing the special relativity?

Solution

With the Galilean relativity, physics laws in classical mechanics are invariant but the Maxwell equations are not invariant. Since the Maxwell equations have been proved to be correct experimentally. The Galilean relativity is therefore incorrect for the electromagnetism and the correct relativity, the special relativity, was developed for the invariance of both classical mechanics and electromagnetism.

2. Answer each of the following questions in **one sentence** without using any mathematical formulas. Your one sentence should answer or define the basic physics or mathematics of each question. **(a)** What is the Galilean transformation? **(b)** What is the Lorentz transformation?

Solution

(a) (2 points) The Galilean transformation is the transformation between inertial frames that keeps the scalars in the Euclidian space invariant.

(b) (2 points) The Lorentz transformation is the transformation between inertial frames that keeps the scalars in the Minkowski space invariant.

3. (4 points) Two basic postulates of the theory of special relativity are usually listed in many book as (I) the laws of nature are identical in all inertial frames; and (II) the speed of light in vacuum is a constant to all observers in all inertial frames. The theory of special relativity can be derived from these two postulates. Can you replace the 2nd postulate with a completely difference mathematical statement? Explain why the one with the constant speed of light is equivalent with the one you replace.

Solution

The 2nd postulate can be replaced (**2 points**) by the statement of "The space consistent with the Einstein relativity is four-dimensional Minkowski space and the scalars in the Minkowski space is invariant under transformations between the inertial frames." (**2 points**) The Lorentz transformation is the transformation that preserves the scalar in the Minkowski space and the space of light is a constant under the Lorentz transformation and therefore a consequence of the invariance of the scalar in the Minkowski space.

4(a) (2 points) Show that the speed of light is the maximal speed in the special relativity.

Solution

From the Lecture Note, the Lorentz transformation for the velocity is

$$u_{\parallel} = \frac{v + u'_{\parallel}}{1 + \vec{v} \cdot \vec{u}\,'/c^2} \qquad \text{and} \qquad \vec{u}_{\perp} = \frac{\vec{u}\,'_{\perp}}{\gamma_v (1 + \vec{v} \cdot \vec{u}\,'/c^2)}.$$

If $v = c$ and $u' = c$, $u_{\parallel} = c$, $\vec{u}_{\perp} = 0$, and $u = c$. Therefore, the maximal speed of an object is c no matter where you observe it.

4(b) (2 points) Explain the reason that a particle traveling with the speed of light has to be massless.

Solution

For a particle traveling with $v = c$, $\gamma = \infty$. Since the energy of a particle is $E = \gamma mc^2$, $m = 0$ is necessary for a finite energy of a particle with $\gamma = \infty$.

5(a) (3 points) Explain from physics point view why Lorentz transformations have to be a linear transformation between two inertial reference frames.

Solution

Since the space is homogeneous, the origin of a frame is arbitrary. The transformation should not depend on the location of the origin, even with a change of the origin. Consider a linear transformation between K and K'

$$x' = c_{11}x + c_{12}t \qquad \text{and} \qquad t' = c_{21}x + c_{22}t$$

where c_{ij} are independent of $(x,\, t)$. For a change of the origins $(x, t) \to (x - x_0,\, t - t_0)$ and $(x', t') \to (x' - x'_0,\, t' - t'_0)$, the transformation becomes

$$x' - x'_0 = c_{11}(x - x_0) + c_{12}(t - t_0) \qquad \text{and} \qquad t' - t'_0 = c_{21}(x - x_0) + c_{22}(t - t_0)$$

which is the same transformation that is independent of the origin

$$x'_0 = c_{11}x_0 + c_{12}t_0 \qquad \text{and} \qquad t'_0 = c_{21}x_0 + c_{22}t_0$$

Therefore, a linear transformation is independent of the choose of the origin. Nonlinear transformations depend on the choose of the origin and, therefore, the space is not homogeneous anymore. For example, for a nonlinear transformation of

$$x' = f_1(x, t) \qquad \text{and} \qquad t' = f_2(x, t)$$

the transformation becomes

$$x' - x'_0 = f_1(x - x_0, t - t_0) \qquad \text{and} \qquad t' - t'_0 = f_2(x - x_0, t - t_0)$$

after a change of the origins. A nonlinear transformation depends on where the origin is.

5(b) (3 points) Explain from mathematical point view why Lorentz transformations can be represented by six infinitesimal generators.

Solution

The Lorentz transformation is a transformation in four-dimensional space that satisfies condition

$$\mathbf{A}^T \mathbf{g} \mathbf{A} = \mathbf{g}$$

which yields $4 \times 4 = 16$ equations for 16 elements of the Lorentz transformation $\mathbf{A}$. Since $\mathbf{A}^T\mathbf{g}\mathbf{A} = (\mathbf{A}^T\mathbf{g}\mathbf{A})^T$ is a symmetric matrix, out of these 16 equations, there are 10 independent equations for the elements of $\mathbf{A}$. There are therefore $16 - 10 = 6$ independent parameters for the Lorentz transformation.

5(c) (2 points) Explain from physics point view why Lorentz transformations have six infinitesimal generators.

Solution

A general Lorentz transformation could be a combination of transformations between different inertial frames that move in three orthogonal directions, (x, y, z), and a rotation in the three-dimensional configuration space that can be constructed using three basic rotational transformation along three orthogonal rotational axes. Therefore, there are six basics transformations for a general Lorentz transformation.

6. Two cosmic events observed from the Earth occur at (ct_1, x_1) and (ct_2, x_2), where the distance between the two events in the Minkowski space satisfies

$$c^2(t_1 - t_2)^2 - (x_1 - x_2)^2 < 0 \qquad \text{and} \qquad x_2 > x_1$$

6(a) (3 points) Is it possible to have an inertial reference frame K' in which these two events are observed at (ct'_1, x'_1) and (ct'_2, x'_2) with

$$c^2(t'_1 - t'_2)^2 - (x'_1 - x'_2)^2 < 0 \qquad \text{and} \qquad x'_2 < x'_1$$

where (ct'_1, x'_1) and (ct'_2, x'_2) are the Lorentz transformations of (ct_1, x_1) and (ct_2, x_2), respectively. You need to explain your reasoning.

Solution

The two events of $ds^2 < 0$ with $x_2 > x_1$ and $x'_2 < x'_1$ are at two different sides of $x = 0$ line in the Minkowski space and are separated by the light-like interval of $ds^2 = 0$ in the Minkowski space. Since the Lorentz transformation is a continuous transformation, it cannot transfer any event from one side of an interval to the another side of the interval by crossing the light-like interval. Therefore, it is not possible to transfer these two events using any Lorentz transformation from one from another.

6(b) (3 points) Discuss any significance or consequence of your answer in **6(a)** to any possible events occurred in our universe.

Solution

If the two events of $ds^2 < 0$ with $x_2 > x_1$ and $x'_2 < x'_1$ could be transferred from each other by a Lorentz transformation, the orientation (the front and back sides) of any object could be altered. The Lorentz transformation is consistent with the homogeneity and isotropy of the Minkowski space and the orientation of any object cannot be altered by the Lorentz transformation.